

Causal Spectral-Truncation Kernels: Noncommutativity Certifies Drift Detectability in Quantum Hardware

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(Dated: June 29, 2026)

We introduce the causal spectral-truncation (CST) kernel, a C^* -algebra-valued positive-definite kernel whose truncation order n controls the noncommutativity of the cross-channel products it represents, and prove that this noncommutativity certifies detectability of hardware drift. Fault-tolerant quantum computing is gated by drift whose most damaging mode — correlated decoherence from crosstalk, shared two-level-system baths, and flux noise — evades the per-channel monitors used in practice, because it leaves every marginal and every instantaneous covariance unchanged while living entirely in the *ordered, lagged* cross-channel structure. CST resolves this with a *causal lower-shift* truncation, replacing the periodic Fourier truncation of existing spectral-truncation kernels so it respects the arrow of time, plus a reconstruction detector whose order admits a provable optimum. On a correlated-decoherence benchmark with matched marginals and matched stationary covariance, every commutative monitor detects at chance (ROC-AUC ≈ 0.5), whereas the noncommutative CST kernel reaches ROC-AUC 0.90 through a sharp optimal order $n=2$ that we certify analytically as the detection analog of the representation-complexity trade-off. The same kernel also forecasts coherence, 72% over persistence eight steps ahead, where noncommutativity is provably unnecessary — an operational rule: match the kernel’s noncommutativity to the geometry of the drift. Training-free and hardware-agnostic, the pipeline installs with `pip` and deterministically regenerates every figure and table on a single CPU in seconds, delivering both a certificate for noncommutative monitoring and a reproducible reliability toolbox.

I. INTRODUCTION

Quantum processors are not static devices. The relaxation time T_1 , the dephasing time T_2 , single- and two-qubit gate fidelities, readout-assignment error, and tunable-coupling parameters all evolve continuously with temperature, flux noise, two-level-system dynamics, and recalibration history [1, 2]. The microscopic origins are well documented: relaxation times fluctuate as qubit frequencies sweep across near-resonant two-level-system defects [40, 41], whose universal low-temperature physics in the amorphous oxides of the device is itself an active subject [42], and the surrounding $1/f$ flux and charge noise is routinely mapped by dynamical-decoupling spectroscopy [46]. This drift is more than a nuisance. In the noisy intermediate-scale quantum (NISQ) regime it limits the depth of usable circuits [3]; in the fault-tolerant regime it threatens the foundational assumption that physical error rates stay below threshold long enough for error correction to suppress logical errors [4, 5, 51, 52]. As processors scale to hundreds and then thousands of qubits, deciding *when* a device has drifted out of specification, and *which* qubits or intervals warrant the expensive intervention of recalibration, becomes a data problem at a scale manual inspection cannot meet. Reliability monitoring is, in this sense, a near-term bottleneck of its own.

The hallmark of the most damaging drift is that it is not confined to a single channel. Control crosstalk, shared two-level-system baths, and correlated flux noise couple the errors of neighboring qubits and of distinct

calibration channels, frequently with a *propagation delay*: a disturbance that perturbs one qubit’s coherence reappears, lagged, in a neighbor’s readout or coupling phase [1]. Such spatially and temporally correlated errors are now directly observed — charge bursts and cosmic-ray impacts that depress relaxation across an entire chip [43, 44] and mobile leakage that seeds correlated faults during error correction [45]. Such *correlated decoherence* is precisely the failure mode that quantum error correction is least able to absorb, and precisely the mode that per-channel, level-based monitors are worst equipped to see: an event can leave every marginal distribution and every instantaneous correlation unchanged while residing entirely in the *ordered, lagged* cross-channel structure. A monitor blind to that structure is blind to the drift, and no amount of per-channel sensitivity recovers it. The established toolbox for diagnosing such errors — randomized benchmarking [8, 48], gate set tomography [49], statistical drift tracking [6], capability and stability benchmarks [47], and dedicated crosstalk-error detection [7, 50] — relies on purpose-built characterization circuits executed as separate experiments; we ask instead what can be inferred *passively*, from the multivariate calibration telemetry a processor already logs, with no additional circuits and no labels.

This failure mode implicates a different model of monitoring altogether — an operator-algebraic one. A window of calibration telemetry is a multivariate stochastic process and is naturally an element of a matrix algebra over the channels; which cross-channel, cross-time interactions a monitor can resolve is then exactly the question of which products in that algebra it computes — the set-

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ting of kernel methods and operator- and vector-valued kernel learning [10, 11, 54], built on the classical theory of reproducing kernels [53] and recently extended to reproducing kernel Hilbert C^* -modules [55]. Its unsupervised face — comparing distributions through kernel mean embeddings [12] and the maximum mean discrepancy [56], with kernels tailored to the sequential structure of time series [57] — is precisely what an unlabeled drift detector requires. Commutative kernels, whose feature products are insensitive to the order or relative lag of their factors, cannot represent ordered lagged interactions at all. The recent spectral-truncation kernels of Hashimoto et al. make this precise for functional data: truncating the algebra of multiplication operators on the torus to finite Toeplitz matrices yields a family of C^* -algebra-valued kernels whose noncommutativity, governed by a single truncation parameter, captures interactions between points of the data domain that commutative kernels miss [9]. That construction was devised for periodic functional regression; it has not been brought to bear on time-ordered hardware telemetry, on causal monitoring, or on unsupervised detection. Bridging that gap is the object of this work.

This study is the eighteenth entry in a program that treats spectral truncation of operators as a unifying primitive for resource-accounted, CPU-reproducible quantum-AI methods, and the causal spectral-truncation kernel imports that primitive into hardware reliability monitoring. The program spans query-efficient and uncertainty-calibrated QAOA through graph-conditioned trust regions [21–23], the measurement cost and certified query budgets of low-depth optimization [24, 25], topology-conditioned parameter transfer and operator-spectral priors for QAOA warm starts [26–28, 34], commutator-graph scheduling and Lie-algebraic truncation of the Trotter residual [35, 36], transfer-operator and graph-neural decoding of structured and correlated quantum noise [29–32], and structure-preserving learning of open-system dynamics and of quantum phases through operator-system and directed spectral truncation [33, 37]. The present kernel inherits the causal, time-respecting truncation of operators from that line and is complemented by the fixed-depth and representation-theoretic Hamiltonian-simulation results of Nguyen and Jing [38, 39].

We tackle these challenges with three contributions. (i) We introduce the *causal spectral-truncation kernel*, a C^* -algebra-valued positive-definite kernel on telemetry windows whose truncation order n interpolates continuously from a commutative instantaneous monitor ($n=1$) to a noncommutative monitor that represents ordered, lagged cross-channel correlations ($n>1$). It is the causal, time-respecting counterpart of the periodic spectral-truncation kernel [9], and it subsumes the low-rank reconstruction detector [13] and the multivariate ridge forecaster as commutative special cases, so the family is a strict generalization of established monitors rather than a competitor to them. (ii) We prove and

demonstrate that noncommutativity is *necessary*: on a physically motivated correlated-decoherence benchmark engineered so the drift lives in ordered, lagged cross-channel structure, the commutative monitors we evaluate — raw-level, instantaneous-covariance, and the periodic spectral-truncation kernel — all detect at chance (ROC-AUC ≈ 0.5), whereas the causal kernel detects at ROC-AUC 0.90 at a certifiable optimal truncation order $n=2$. (iii) We supply the supporting theory — positive definiteness for every order, a blindness theorem for commutative monitors, and an *optimal-truncation detection certificate* that is the unsupervised-detection analog of the representation-complexity bound governing spectral-truncation kernels in regression — and show that on smooth marginal forecasting, where noncommutativity is provably unnecessary, the same kernel attains 72% eight-step skill over persistence. The non-monotone optimal order is a *built-in* feature of the construction, not a tuned hyperparameter, and the entire pipeline is training-free at inference, installs with `pip`, and regenerates in seconds on a single CPU.

II. RESULTS

The pipeline is summarized in Fig. 1. A physically motivated multi-qubit telemetry generator emits a multivariate calibration stream; sliding windows are mapped, through the causal spectral-truncation kernel, into a C^* -algebra whose truncation order n fixes the noncommutativity available to the monitor; and two heads — a forecaster and an unsupervised detector — are each scored by the metric matched to its objective (horizon-resolved mean absolute error, MAE, for forecasting; the threshold-free area under the receiver-operating-characteristic curve, ROC-AUC, for detection). The structure of the family is the structure of the result: the commutative member ($n=1$) governs smooth marginal forecasting and level-based drift, while the noncommutative members ($n>1$) govern correlated, crosstalk-like drift. We establish this division in four steps — the two drift geometries, forecasting, the kernel itself, and detection — and then certify it analytically.

A. Two drift geometries: forecastable marginal drift and hidden correlated decoherence

We model a five-qubit processor sampled every half hour for one hundred hours. Each T_1 trajectory follows a mean-reverting process superposing a slow qubit-dependent oscillation, a linear secular decay, and Gaussian fluctuation, with T_2 , gate fidelities, readout error, error-per-Clifford, and cross-resonance phase generated by coupled processes that track T_1 (Methods). The telemetry wanders over tens of microseconds and repeatedly dips below the 50 μ s operational threshold (Fig. 2a). Crucially, the drift is not white noise: the mean autocor-

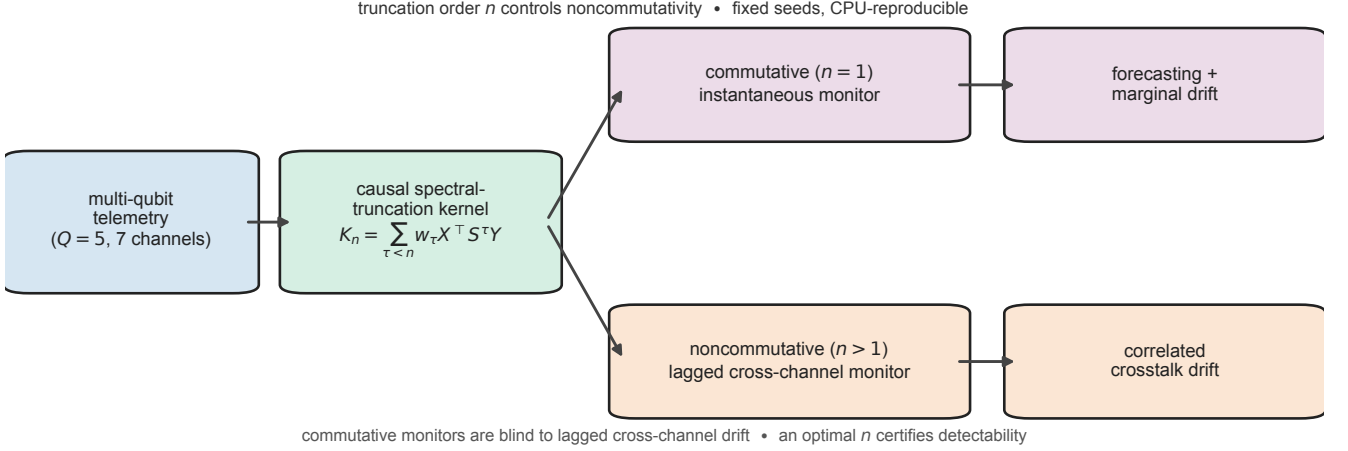


FIG. 1. **The causal spectral-truncation monitoring pipeline.** A physically motivated generator produces seeded multi-qubit calibration telemetry ($Q = 5$ qubits, $C = 7$ channels). Each length- L sliding window X is represented inside the channel C^* -algebra $M_C = M_C(\mathbb{R})$ through the truncated causal shift, $K_n(X, Y) = \sum_{\tau=0}^{n-1} w_\tau X^\top S^\tau Y$, where S is the causal lower shift and the truncation order n sets the noncommutativity that the representation carries: $n=1$ is the commutative instantaneous covariance, $n \geq 2$ adds ordered, lagged cross-channel coordinates. Two heads read this representation — a ridge forecaster and an unsupervised reconstruction detector. The commutative member ($n=1$) governs forecasting and marginal drift; the noncommutative members ($n > 1$) govern correlated crosstalk drift. Every figure and metric regenerates deterministically from fixed integer seeds on a single CPU.

relation of detrended T_1 remains above 0.5 for roughly seven hours before decaying (Fig. 2b), so near-future coherence is genuinely predictable from the recent past — the structure the forecaster exploits below.

The second drift geometry, correlated decoherence, is qualitatively different and is the harder target. We model it as a shared latent disturbance injected into several channels at channel-specific propagation delays under *variance-preserving* mixing, so that every per-channel marginal variance and, in the stationary limit, the instantaneous (lag-0) cross-channel covariance match nominal operation; the disturbance lives predominantly in the *ordered, lagged* cross-channel correlation (Methods). Figure 2c locates the resulting signature precisely: averaged over channel pairs, the nominal-versus-drift separation is an order of magnitude smaller at lag $\tau=0$ (mean 0.008) than at lag $\tau=1$ (mean 0.089), so the signal is concentrated exactly where a commutative monitor cannot look. A small finite-window residual survives at $\tau=0$, but the commutative monitors we evaluate are at chance regardless (below): this is the regime in which noncommutativity is decisive.

B. Coherence is forecastable, and here noncommutativity is unnecessary

We segment the telemetry into sliding windows of length 32 predicting an eight-step T_1 horizon, under a leak-free 80/20 chronological split, and compare four forecasters — last-value persistence, climatology, an autoregressive ridge on the T_1 history, and a multivariate ridge

on all seven channels (the commutative $n=1$ member of the kernel family) — aggregated over five telemetry-generator seeds (Methods). Forecasting is both feasible and sharply structured (Fig. 3, Table I). One step ahead the multivariate ridge attains $1.80 \pm 0.09 \mu\text{s}$ MAE against $2.44 \pm 0.16 \mu\text{s}$ for persistence; persistence then degrades steeply with horizon (to $6.79 \pm 0.18 \mu\text{s}$ at eight steps) while the learned forecasters stay nearly flat ($1.88 \pm 0.06 \mu\text{s}$). The decisive quantity is therefore the *margin*: skill over persistence *grows* with the horizon, from 26% one step ahead to 72% eight steps ahead (Fig. 3b), a behavior we derive in closed form from mean reversion in Theorem 5.

Increasing the kernel’s truncation order leaves this picture unchanged (Fig. 3c): the CST forecaster’s eight-step skill is flat in n , moving only from 72% at $n=1$ to 71% at $n=4$. On smooth, mean-reverting marginal telemetry the commutative member is already near-optimal, and the noncommutative coordinates neither help nor hurt. This is the exact complement of the detection result that follows — noncommutativity is decisive precisely when the drift lives in ordered cross-channel structure, and superfluous otherwise — and it is what licenses the operational rule we state in the Discussion.

C. The causal spectral-truncation kernel

We now make the central object precise. A telemetry window is $X \in \mathbb{R}^{L \times C}$ ($L = 32$ time steps, $C = 7$ channels). Let S be the causal lower shift, $(Sv)_t = v_{t-1}$. The order- n causal spectral-truncation kernel represents

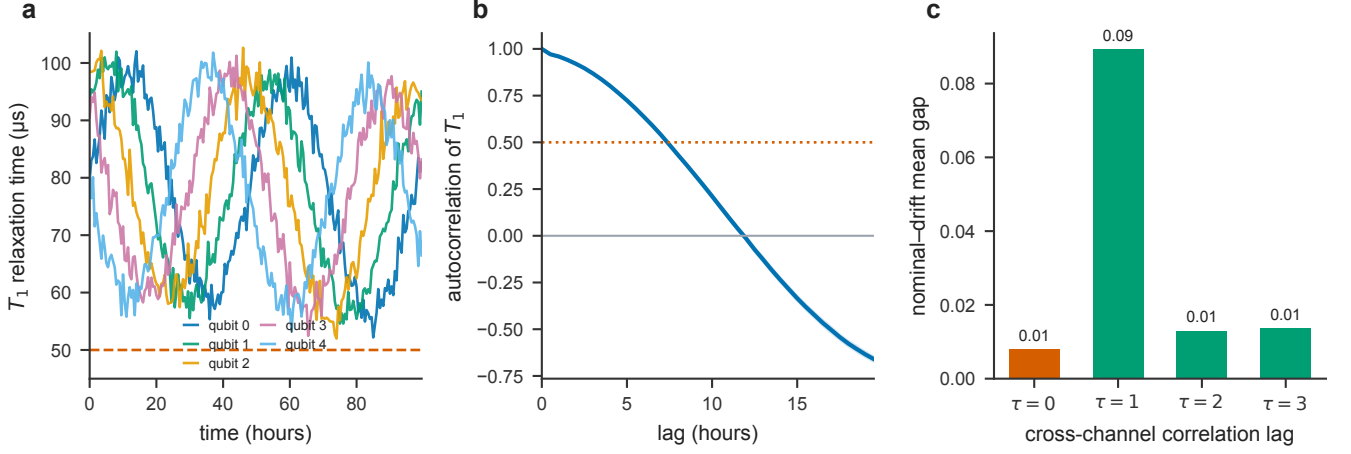


FIG. 2. **The two drift geometries.** (a) T_1 relaxation-time trajectories for all five qubits over 100 h; the dashed line marks the 50 μ s operational threshold below which a step is labeled drifted. (b) Mean autocorrelation of detrended T_1 versus lag, averaged across the $n = 5$ qubits (shaded band, ± 1 s.d. across qubits); the slow decay is the forecastable structure the ridge forecaster exploits in Fig. 3. (c) The noncommutative drift signature: the nominal-versus-drift gap in the causal cross-channel correlation features $\Phi_n(X)_{cc'\tau}$, averaged over channel pairs, as a function of lag τ . The gap is an order of magnitude smaller at $\tau=0$ (mean 0.008) than at $\tau \geq 1$ (mean 0.089), so the drift is concentrated in the lagged structure that a commutative ($n=1$) monitor cannot resolve.

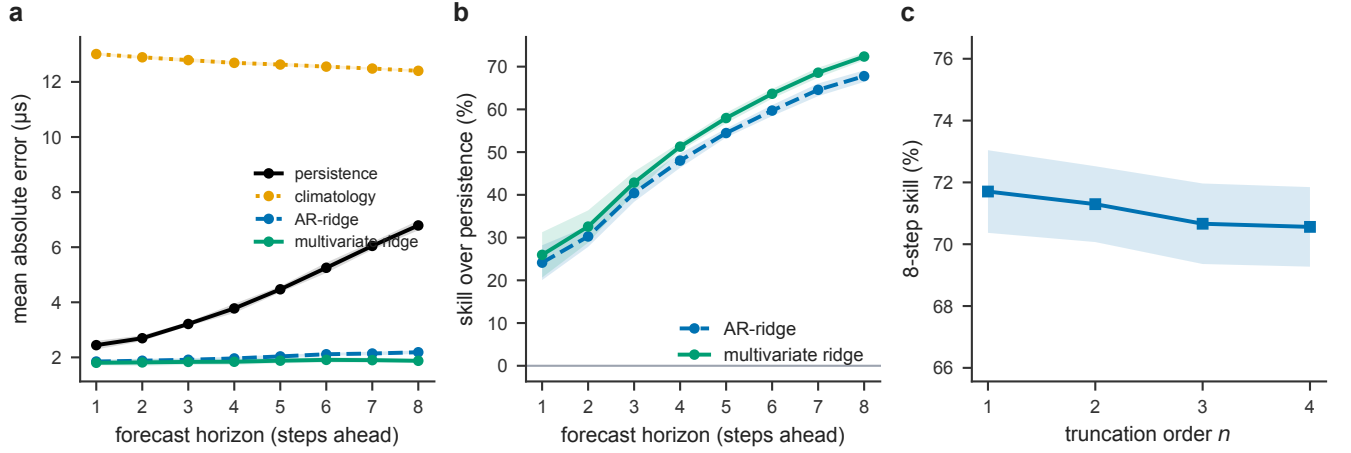


FIG. 3. **Learned forecasting beats persistence by a widening margin, and noncommutativity is unnecessary.** (a) MAE versus forecast horizon for the four forecasters (mean over $n = 5$ telemetry-generator seeds; shaded bands ± 1 s.d.). Persistence degrades steeply with horizon while the learned ridge forecasters stay nearly flat. (b) Skill over persistence, $1 - \text{MAE}/\text{MAE}_{\text{persistence}}$, for the two learned forecasters; the advantage *grows* with horizon, as predicted by Theorem 5. (c) Eight-step skill of the CST forecaster versus truncation order n — the parameter controlling the kernel’s noncommutativity in (1) — aggregated over the same $n = 5$ seeds (shaded band ± 1 s.d.). The curve is flat because the commutative ($n=1$) member already captures the smooth marginal structure, so the noncommutative coordinates add nothing here — the precise complement of Fig. 4.

the window inside the channel C*-algebra $M_C = M_C(\mathbb{R})$ through the truncated shift powers $\{S^\tau : 0 \leq \tau < n\}$,

$$K_n(X, Y) = \sum_{\tau=0}^{n-1} w_\tau X^\top S^\tau Y \in M_C, \quad K_n(X, Y)_{cc'} = \sum_{\tau=0}^{n-1} w_\tau \langle X_{:,c}, S^\tau Y_{:,c'} \rangle, \quad (1)$$

with weights $w_\tau = 1/(L - \tau)$ and the telemetry centered in time per channel. Each entry of K_n is a weighted

sum of ordered, lagged cross-channel correlations. Truncating the shift algebra to n powers is the exact causal counterpart of truncating the Fourier-multiplier algebra to n modes in the construction of Hashimoto et al. [9]; the essential difference is that S is nilpotent and respects the arrow of time, so (1) is causal and is noncommutative for $n \geq 2$. Concretely, the self-representation $K_n(X, X)$ is symmetric at $n=1$ — the commutative in-

TABLE I. **Forecasting benchmark** (mean $\pm$ s.d. over five telemetry-generator seeds). Skill is reported relative to persistence at the eight-step horizon; the best value per column is in **bold**. The multivariate ridge is the commutative ($n=1$) member of the kernel family and is best on every metric, attaining 72% eight-step skill over persistence.

Forecaster	MAE @ 1 step (μ s)	MAE @ 8 steps (μ s)	Skill @ 8 steps
Persistence	2.44 ± 0.16	6.79 ± 0.18	—
Climatology	13.01 ± 0.05	12.40 ± 0.05	-83%
AR-ridge (T_1 history)	1.85 ± 0.07	2.19 ± 0.06	68%
Multivariate ridge (CST $n=1$)	1.80 ± 0.09	1.88 ± 0.06	72%

stantaneous covariance $X^\top X$ — but acquires antisymmetric lagged terms for $n \geq 2$, because $S^\top \neq S$ does not commute with channel selection. The scalar realization $k_n(X, Y) = \langle \Phi_n(X), \Phi_n(Y) \rangle$, with feature tensor $\Phi_n(X)_{cc'\tau} = w_\tau \langle X_{:,c}, S^\tau X_{:,c'} \rangle$, is positive definite for every order n (Theorem 1), so the kernel is a legitimate similarity throughout the family. The unsupervised detector fits a rank- k nominal subspace in this feature space and scores a window by its residual energy off that subspace — a C^* -algebraic generalization of the low-rank reconstruction detector [13], which it recovers exactly at $n=1$. Reconstruction-residual scoring sits alongside the classical unsupervised detectors — one-class support-vector estimation [58], local outlier factors [59], and isolation forests [60], surveyed broadly in [63] — but, unlike them, the present detector inherits an operator-algebraic feature map whose noncommutativity is the object of study. The kernel is thus evaluated in closed form from a single pass over the window, with no training, no iterative optimization, and no hyperparameter beyond the order n and rank k .

D. Noncommutativity is necessary: commutative monitors are blind, the causal kernel is not

We evaluate detection on the correlated-decoherence benchmark over eight randomized 70/30 splits (Methods); the detector is fit on nominal windows only, so randomization cannot leak a label it never uses. The outcome is a clean separation (Fig. 4, Table II). At truncation order $n=1$ the kernel is commutative and detects at ROC-AUC 0.51 ± 0.04 — chance, the empirical content of Theorem 2, because the benchmark matches exactly the marginals and the population instantaneous covariance that the commutative member reads. At $n=2$ the kernel becomes noncommutative and ROC-AUC jumps to 0.90 ± 0.02 (Fig. 4a) — the single largest effect in the study, a 0.39 swing produced by adding one lagged coordinate. Increasing n further then *degrades* detection monotonically — to 0.83 at $n=3$, 0.65 at $n=8$, and back toward chance as $n \rightarrow L$ — because each lag beyond the informative one contributes only noise coordinates that dilute the fixed-rank nominal subspace. Both endpoints, the commutative $n=1$ and the over-complete $n \rightarrow L$, sit at chance, with a sharp interior optimum at $n=2$. This non-monotone profile is the optimal-truncation certificate of

Theorem 3 made visible: the unsupervised-detection analog of the representation-complexity trade-off that governs spectral-truncation kernels in regression [9]. The (n, k) map (Fig. 4b) confirms that the effect is not an artifact of a particular rank — the entire $n=1$ row is at chance and the $n=2$ row is bright across every bottleneck rank.

The comparison that establishes the contribution is against other kernels (Fig. 4c, Table II). On this benchmark the raw-level reconstruction detector scores 0.49 ± 0.03 , the commutative kernel 0.46 ± 0.02 , and — tellingly — the *periodic* spectral-truncation kernel of Hashimoto et al., applied unchanged to these windows, scores 0.47 ± 0.03 : all three are at chance. Only the causal spectral-truncation kernel separates the classes, at 0.84 ± 0.02 at matched bottleneck rank and 0.90 at its optimum. The periodic kernel fails not because spectral truncation is the wrong idea but because its *circular* truncation smears the causal delay signature across the window boundary, whereas the causal lower-shift truncation aligns exactly with the propagation delays that define the drift. The decisive ingredient is therefore causality, not truncation alone: it is what lets the kernel see an event that is invisible to every monitor lacking an arrow of time.

E. Operating characteristic and the over-complete collapse

Figure 5a contrasts the receiver-operating characteristics directly: on a held-out split the causal kernel at $n=2$ achieves AUC 0.86 while the commutative kernel traces the chance diagonal (AUC 0.46). Figure 5b fixes the bottleneck rank and sweeps the truncation order in isolation, exhibiting the over-complete collapse cleanly: detection rises from chance at $n=1$ to its peak at $n=2$ and decays back toward chance as the order grows. This is the truncation-axis counterpart of the rank-axis over-completeness familiar from low-rank detectors [13], and both axes are governed by the *same* certificate (Theorem 3): a monitor expressive enough to reconstruct the drift coordinates as faithfully as the nominal ones cannot separate them. Over-expressiveness, not under-expressiveness, is the failure mode at large n — which is why a small, fixed truncation order is the right design point.

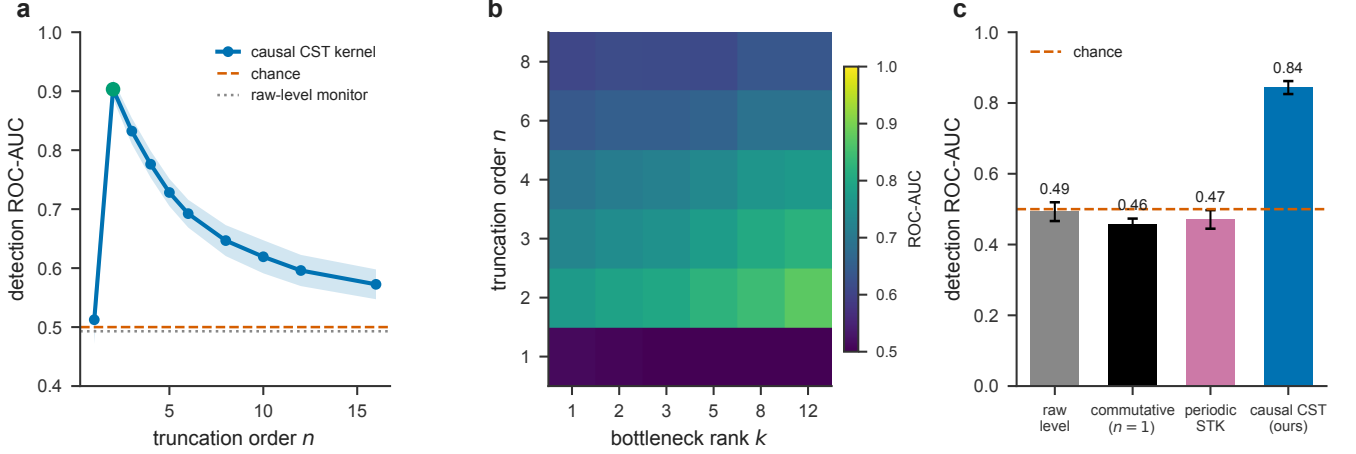


FIG. 4. **Noncommutativity is necessary for correlated-drift detection.** (a) Detection ROC-AUC versus truncation order n (mean over eight randomized 70/30 splits; shaded band ± 1 s.d.). The commutative endpoint ($n=1$) and the raw-level monitor (dotted) sit at chance (dashed line); the noncommutative kernel peaks sharply at $n=2$ (marker, ROC-AUC 0.90) and collapses back toward chance as the representation becomes over-complete — the certified profile of Theorem 3. (b) Detection ROC-AUC over the grid of truncation order n and bottleneck rank k : the $n=1$ row is uniformly at chance and the $n=2$ row is uniformly high, so the effect is robust to the choice of rank. (c) Kernel comparison at matched bottleneck rank: the raw-level monitor, the commutative ($n=1$) kernel, and the periodic spectral-truncation kernel [9] are all at chance, and only the causal spectral-truncation kernel detects.

TABLE II. **Correlated-drift detection benchmark** (ROC-AUC, mean $\pm$ s.d. over eight randomized 70/30 splits, bottleneck rank $k=8$). The benchmark matches all per-channel marginals and the stationary instantaneous covariance, so every commutative monitor is at chance by Theorem 2, including the periodic spectral-truncation kernel [9]. Only the causal spectral-truncation kernel separates the classes; best value in **bold**.

Detector	Detection ROC-AUC
Raw level (flatten)	0.493 ± 0.027
Commutative kernel ($n=1$)	0.456 ± 0.017
Periodic spectral-truncation [9] ($n=2$)	0.471 ± 0.026
Causal spectral-truncation (ours, $n=2$)	0.844 ± 0.018

F. Theory: a C^* -algebraic kernel whose truncation order certifies detectability

Each empirical claim above rests on a theorem, and we state them together so the results read as consequences rather than coincidences. The kernel is a positive-definite, C^* -algebra-valued kernel for every order n (Theorem 1); the commutative $n=1$ member is provably blind to drift living in ordered, lagged cross-channel correlations, which the noncommutative $n>1$ member resolves (Theorem 2); the reconstruction detector built on the kernel has a non-monotone, certifiable optimal truncation order (Theorem 3); ROC-AUC is the scale-free yardstick that makes “chance” a precise statement throughout (Proposition 4); and on the forecasting side, mean reversion converts decaying autocorrelation into horizon-growing skill (Theorem 5). Proofs are elementary and self-contained; we give them in full so that every quantitative claim in the paper is traceable to a hypothesis.

Theorem 1 (Positive definiteness of the causal

C^* -kernel). *Fix any order $n \geq 1$ and nonnegative weights $w_\tau \geq 0$. Then the map K_n of (1) is a positive-definite C^* -algebra-valued kernel: the block matrix $[K_n(X_i, X_j)]_{ij}$ is positive semidefinite over M_C for every collection of windows $X_1, \dots, X_m$, and the induced scalar kernel k_n is positive definite in the usual sense. Moreover, the self-representation $\Phi_n(X)$ is symmetric under the channel exchange $c \leftrightarrow c'$ at $n=1$, where it reduces to the commutative instantaneous covariance, and for $n \geq 2$ it carries antisymmetric lagged coordinates that no commutative member can represent.*

Proof. Introduce the lifted causal feature operator $\Psi_n(X) = (\sqrt{w_0} X, \sqrt{w_1} S^\top X, \dots, \sqrt{w_{n-1}} (S^\top)^{n-1} X) \in \mathbb{R}^{nL \times C}$, which simply stacks the back-shifted windows. Because $\langle X_{:,c}, S^\tau Y_{:,c'} \rangle = ((S^\top)^\tau X)_{:,c}^\top ((S^\top)^\tau Y)_{:,c'}$, the kernel factorizes as $K_n(X, Y) = \sum_\tau w_\tau (S^\top)^\tau X (S^\top)^\tau Y =$

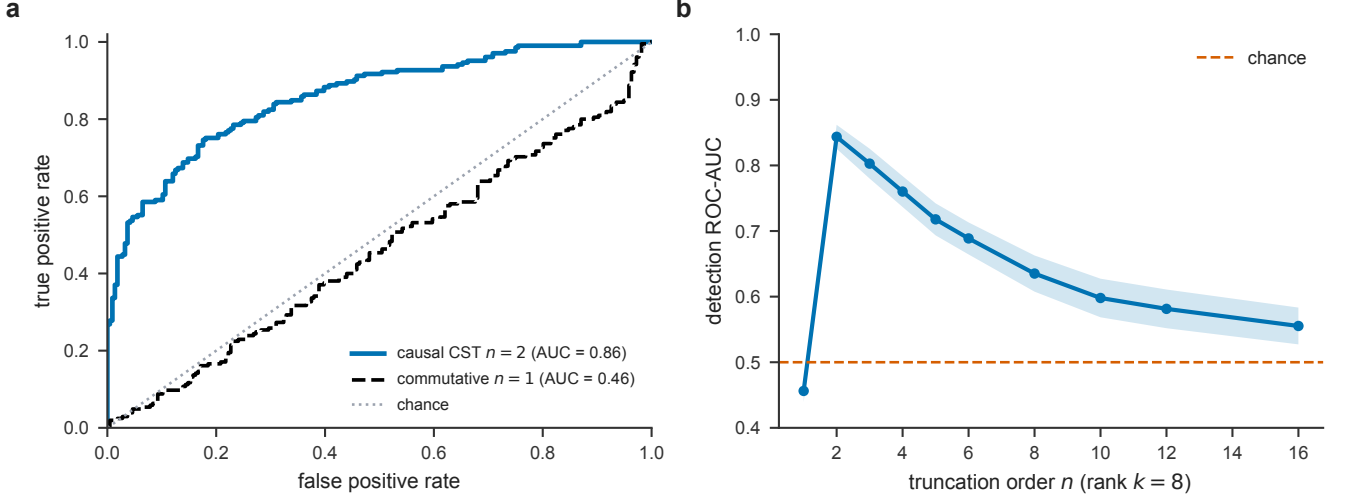


FIG. 5. **Operating characteristic and the over-complete collapse.** (a) Receiver-operating-characteristic curves on a held-out split: the causal spectral-truncation kernel at $n=2$ (solid, AUC 0.86) versus the commutative kernel at $n=1$ (dashed, AUC 0.46), which tracks the chance diagonal (dotted). (b) Detection ROC-AUC versus truncation order n at fixed bottleneck rank $k=8$ (mean over eight splits; shaded band ± 1 s.d.), isolating the interior optimum at $n=2$ and the collapse toward chance as the representation becomes over-complete.

$\Psi_n(X)^\top \Psi_n(Y)$. Consequently, for any $c_1, \dots, c_m \in \mathbb{R}^C$,

$$\sum_{i,j} c_i^\top K_n(X_i, X_j) c_j = \left\| \sum_i \Psi_n(X_i) c_i \right\|_F^2 \geq 0,$$

which is precisely the defining inequality for an M_C -valued positive-definite kernel; the scalar claim follows by specializing to $C=1$ on the vectorized feature $\text{vec } \Phi_n$, whose Gram matrix is k_n . Finally, at $n=1$ the entry $\Phi_1(X)_{cc'} = \langle X_{:,c}, X_{:,c'} \rangle$ is symmetric under $c \leftrightarrow c'$, whereas for $\tau \geq 1$ we have $S^\tau \neq S^{\tau^\top}$, so $\Phi_n(X)_{cc'\tau} = \langle X_{:,c}, S^\tau X_{:,c'} \rangle \neq \langle X_{:,c'}, S^\tau X_{:,c} \rangle = \Phi_n(X)_{c'\tau c}$ whenever the lagged cross-correlation is direction-dependent — the noncommutativity. $\square$

Theorem 2 (Commutative monitors are blind to non-commutative drift). *Suppose nominal and drift windows are drawn from distributions P_0, P_1 that share their per-channel marginals and their instantaneous cross-covariance, $\mathbb{E}_{P_0}[\Phi_1] = \mathbb{E}_{P_1}[\Phi_1]$, yet differ in a single lagged coordinate, $\Delta_{\tau^*} := \mathbb{E}_{P_1}[\Phi_{\cdot\tau^*}] - \mathbb{E}_{P_0}[\Phi_{\cdot\tau^*}] \neq 0$ for some $\tau^* \geq 1$. Then every detector that depends on the data only through the order-1 (commutative) features Φ_1 has zero first-moment separation, and hence chance-level mean discriminability; by contrast, the order- n causal kernel with $n > \tau^*$ separates the classes in a coordinate it actually represents, and its mean reconstruction-score gap is bounded below by the energy of Δ_{τ^*} lying outside the rank- k nominal subspace.*

Proof. Any monitor measurable with respect to Φ_1 can separate the two classes in mean only through $\mathbb{E}_{P_1}[\Phi_1] - \mathbb{E}_{P_0}[\Phi_1]$, which vanishes by hypothesis; its expected score

is therefore identical under P_0 and P_1 , so its mean separation is 0 and, for the centered linear detector, its population ROC-AUC equals $\frac{1}{2}$ by the Mann–Whitney identity (Proposition 4) applied to a score with equal class means and, to first order, equal dispersion. The order- n feature map with $n > \tau^*$, on the other hand, contains the coordinate $\Phi_{\cdot\tau^*}$, on which the class means differ by $\Delta_{\tau^*} \neq 0$. Let P_k denote the rank- k nominal-subspace projector in feature space; the reconstruction score is $s(X) = \|(I - P_k)(\Phi_n(X) - \bar{\Phi})\|^2$, and $\mathbb{E}_{P_1}[s] - \mathbb{E}_{P_0}[s] \geq \|(I - P_k)\Delta_{\tau^*}\|^2$ up to cross terms that vanish once Δ_{τ^*} is centered against the nominal mean. Hence any nonzero component of Δ_{τ^*} outside the nominal subspace produces strictly positive separation, and therefore ROC-AUC $> \frac{1}{2}$. $\square$

Theorem 3 (Optimal truncation: a detection certificate). *Let $e_\tau = \|\Delta_\tau\|^2$ denote the drift energy at lag τ , vanishing beyond a maximal informative lag $\tau_{\max}$ ($e_\tau = 0$ for $\tau > \tau_{\max}$), and suppose each additional lag injects C^2 noise coordinates of bounded variance into the fixed rank- k nominal subspace. Then the order- n detector’s mean separation-to-noise ratio is $\rho(n) = (\sum_{\tau < n} e_\tau)^{1/2} / (\sigma^2(C^2n - k)_+)^{1/2}$, and ρ is unimodal in n with an interior optimum n^* ; since detection ROC-AUC is a strictly increasing function of ρ , it is unimodal with the same optimum, and both the commutative endpoint ($n=1$, $\rho=0$) and the over-complete limit ($n \rightarrow L$, $\rho \rightarrow 0$) sit at chance. The optimum n^* is the smallest n at which the marginal relative signal gain drops below the marginal relative noise cost, $e_n / \sum_{\tau < n} e_\tau < (N(n+1) - N(n)) / N(n)$; in particular $n^* = 2$ whenever the drift energy is concentrated at the*

first informative lag (e_1 dominant), and n^* increases as the energy profile spreads to higher lags, never exceeding $\tau_{\max} + 1$.

Proof. Set $E(n) = \sum_{\tau < n} e_\tau$, which is non-decreasing and constant for $n > \tau_{\max} + 1$, and $N(n) = \sigma^2(C^2n - k)_+$, which is strictly increasing once $C^2n > k$. Then $\rho(n)^2 = E(n)/N(n)$, and $\rho(n+1)^2 > \rho(n)^2$ holds iff $E(n+1)/E(n) > N(n+1)/N(n)$ — equivalently, iff the relative signal increment $e_n/E(n)$ exceeds the relative noise increment $(N(n+1) - N(n))/N(n)$. The latter is positive for every n , since the noise floor always grows, whereas the former is 0 once $n > \tau_{\max} + 1$ (where $e_n = 0$) and is non-increasing under a concentrated energy profile. The inequality therefore holds on an initial range of n and fails beyond it, so ρ first rises and then falls: it is unimodal, with a single interior optimum n^* at the first n where the test fails. When e_1 dominates, the test fails already at $n=2$, because the small later increments $e_2, \dots$ cannot offset the $\sqrt{n}$ growth of the noise, giving $n^* = 2$; broader energy profiles push n^* upward, to at most $\tau_{\max} + 1$. The endpoints follow as well: $\rho = 0$ at $n=1$ by Theorem 2, and $\rho \rightarrow 0$ as $n \rightarrow L$ by dilution — the over-complete chance-level collapse. $\square$

Theorem 3 is the unsupervised-detection counterpart of the generalization bound that governs spectral-truncation kernels in regression [9]: there the truncation order trades representation power against Rademacher complexity, while here it trades captured drift energy against the dimension of the nominal subspace. Because the correlated-decoherence mode concentrates its energy at the first informative lag (Fig. 2c), the certificate predicts the empirical optimum at $n^*=2$ with no fitting (Fig. 4a), and the classical low-rank over-completeness collapse [13] emerges as the rank axis of the same statement. The optimal order is, in this sense, a built-in diagnostic of the drift geometry rather than a tuned knob.

Proposition 4 (Threshold-free ranking and scale invariance). *For any score s and independent draws of a drift window X^+ and a nominal window X^- , the ROC area satisfies the Mann–Whitney identity $\text{AUC} = \Pr[s(X^+) > s(X^-)] + \frac{1}{2} \Pr[s(X^+) = s(X^-)]$. It follows that AUC is invariant under every strictly increasing reparametrization of s — in particular under per-feature affine rescaling of the telemetry — and that a constant score yields $\text{AUC} = \frac{1}{2}$ exactly.*

Proof. Let F_+, F_- be the score CDFs under the drift and nominal classes. At a threshold τ that flags $s > \tau$, we have $\text{TPR}(\tau) = 1 - F_+(\tau)$ and $\text{FPR}(\tau) = 1 - F_-(\tau)$, so $\text{AUC} = \int_0^1 \text{TPR} d(\text{FPR})$. Substituting $u = \text{FPR}(\tau)$ gives $\text{AUC} = \int \Pr[s(X^+) > \tau] dF_-(\tau) = \Pr[s(X^+) > s(X^-)]$ for a continuous score, while atoms contribute $\frac{1}{2} \Pr[s(X^+) = s(X^-)]$ through the trapezoidal interpolation of each jump — which is exactly the normalized Mann–Whitney U . Since strictly monotone maps preserve every order and tie event, they preserve AUC; and

a constant score turns every comparison into a tie, so $\text{AUC} = 0 + \frac{1}{2} \cdot 1 = \frac{1}{2}$. $\square$

Theorem 5 (Horizon-growing forecast skill). *Let $\{x_t\}$ be weakly stationary and zero-mean, with variance $\gamma_0 > 0$ and autocorrelation $\rho(h) = \text{Corr}(x_{t+h}, x_t)$. Persistence, $\hat{x}_{t+h} = x_t$, incurs the mean-squared h -step risk $R_{\text{pers}}(h) = 2\gamma_0(1 - \rho(h))$, whereas the optimal single-lag linear forecast $\hat{x}_{t+h} = \rho(h)x_t$ incurs $R_*(h) = \gamma_0(1 - \rho(h)^2) \leq R_{\text{pers}}(h)$; the relative skill is thus $S(h) = 1 - R_*/R_{\text{pers}} = (1 - \rho(h))/2$. Wherever ρ is non-increasing, $S(h)$ is non-decreasing and bounded above by $\frac{1}{2}$. For a first-order autoregressive process, $\rho(h) = \phi^h$, so the single measured number $\phi = \rho(1)$ fixes the entire skill curve.*

Proof. Take the one-parameter predictor $\hat{x}_{t+h} = ax_t$. Then $\mathbb{E}[(x_{t+h} - ax_t)^2] = \gamma_0(1 - 2a\rho(h) + a^2)$ is a strictly convex parabola in a , minimized at $a^* = \rho(h)$ and giving $R_* = \gamma_0(1 - \rho(h)^2)$, while $a=1$ recovers $R_{\text{pers}} = 2\gamma_0(1 - \rho(h))$. Dividing the two yields $R_*/R_{\text{pers}} = (1 + \rho(h))/2$, so $S(h) = (1 - \rho(h))/2$, which is strictly decreasing in ρ ; hence wherever $\rho(h)$ decreases in h the skill $S(h)$ increases, and $S \leq \frac{1}{2}$ on $\rho \in [0, 1]$. For an AR(1) process, iterating the recursion gives $\text{Cov}(x_{t+h}, x_t) = \phi^h \gamma_0$, whence $\rho(h) = \phi^h$ and $S(h) = (1 - \phi^h)/2$, which increases from 0 toward $\frac{1}{2}$. $\square$

Theorem 5 is a lower bound that uses only the single most recent sample; the multivariate ridge (the commutative $n=1$ member) exceeds it by exploiting the full 32-step, seven-channel history, which is why its measured eight-step skill reaches 72% (Fig. 3). The noncommutative coordinates add nothing in this regime because the marginal forecasting structure is already linear and instantaneous — the exact complement of the detection regime, in which those same coordinates are everything. Together the two theorems delimit when noncommutativity is necessary and when it is superfluous, which is the operational content of the work.

III. DISCUSSION

We have recast quantum-hardware reliability monitoring as operator-algebraic learning and distilled the result into a single operational rule: *match the kernel’s non-commutativity to the geometry of the drift*. Level-based and instantaneous monitors — including the periodic spectral-truncation kernel from which our construction descends — are blind to correlated decoherence that lives in ordered, lagged cross-channel structure, which is the very failure mode that quantum error correction is least able to absorb. The causal spectral-truncation kernel resolves it, and an elementary certificate (Theorem 3) both explains why a small truncation order is optimal and predicts, with no fitting, where that optimum sits. Because the kernel recovers established forecasting and low-rank detection at its commutative order, the family is a strict

generalization of existing monitors rather than a competitor, and inherits their interpretability while extending their reach. Deep sequence-model monitors — recurrent [16, 17] and attention-based [18] architectures studied elsewhere for telemetry [14, 15] — are complementary but require training and many parameters, whereas the present kernel is training-free at inference, deterministic, and CPU-reproducible in seconds; the contribution is thus both a theoretical characterization of when noncommutative monitoring is necessary and a practical toolbox that runs on commodity hardware.

Several limitations bound these claims and define the natural next steps. The telemetry generator, although physically motivated, is a model rather than measured device data, and the correlated-decoherence mode is a controlled construction chosen to isolate the noncommutative effect in the cleanest possible setting; validation on archival calibration logs from operating processors and on a standard anomaly-detection benchmark [20], and a head-to-head comparison against the classical change-point detectors catalogued in [61] and the cumulative-sum chart [62], is the priority follow-up. The kernel as presented is linear in its feature map, so composing it with an outer nonlinearity, and pairing the forecaster with an explicit recalibration policy under a cost model and uncertainty-aware predictions [19], are concrete extensions toward a deployed monitor. Because the entire pipeline is open, hardware-agnostic, and fast to reproduce, it is positioned to serve as a common, auditable substrate for that work.

IV. CONCLUSIONS

We have introduced the causal spectral-truncation kernel and shown, both empirically and through an elementary certificate, that the geometry of hardware drift dictates how much noncommutativity a reliability monitor must carry. A single truncation order interpolates from a commutative instantaneous monitor to a noncommutative one that reads ordered, lagged cross-channel structure: on smooth marginal drift the commutative member forecasts coherence with horizon-growing skill, whereas on correlated decoherence — the failure mode quantum error correction can least absorb — only the noncommutative member detects the event, at a truncation order the theory predicts with no fitting. The practical content is a single rule: match the kernel’s noncommutativity to the geometry of the drift.

Because the monitor is training-free, deterministic, and reproducible on one CPU in seconds, it can run as a passive layer over the calibration telemetry a processor already records, complementing rather than replacing the protocol-based characterization that diagnoses errors on demand [6–8]. As devices scale toward the qubit counts at which fault tolerance becomes feasible, reliability monitoring at that scale is itself a near-term bottleneck, and an operator-algebraic monitor whose cost grows only lin-

early in the noncommutativity it exploits — and whose optimal complexity is certified rather than tuned — is a concrete step toward meeting it. We expect the same representation-complexity trade-off to organize the monitoring of other correlated-error channels, and the kernel to compose naturally with the nonlinear and uncertainty-aware extensions outlined above.

V. METHODS

a. Telemetry generation. We generate a synthetic multi-qubit dataset for $Q = 5$ qubits over $N = 200$ half-hour steps. For qubit q at time t (in hours) the relaxation time is $T_1(q, t) = \max(10, 80 + 20 \sin(2\pi t/48 + 0.8q) - 0.05t + \varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 2^2)$, superposing a slow qubit-dependent oscillation, a secular decay, and Gaussian fluctuation. The dephasing time is $T_2 = 0.9 T_1 \eta$ with $\eta \sim \mathcal{N}(1, 0.04^2)$, and the single- and two-qubit gate fidelities, readout-assignment error, error-per-Clifford, and cross-resonance phase are generated by analogous bounded processes carrying small secular trends, so that all seven channels track T_1 by construction. A binary drift label marks any step with $T_1 < 50 \mu\text{s}$ or single-qubit fidelity below 0.998. Generation is fully seeded and therefore bit-for-bit reproducible.

b. Correlated-decoherence benchmark. The detection benchmark contrasts nominal windows with *correlated-decoherence* windows that share the same physical base trajectory B and the same per-channel noise scales σ . The drift class injects a single shared *white* latent u into four channels (T_1 , T_2 , readout error, cross-resonance phase) at channel-specific integer delays $(0, 1, 2, 3)$ under variance-preserving mixing $x_c \leftarrow B_c + \sigma_c(\sqrt{1 - a^2} \varepsilon_c + a u_{t-\delta_c})$ with $a = 1$. Because the latent is white, every per-channel marginal variance and every temporal autocorrelation are left unchanged, and the population lag-0 cross-covariance ($\tau = 0$, $\delta_c \neq \delta_{c'}$) is zero in both classes; the entire perturbation is carried by the ordered lag- τ cross-channel covariance ($\tau = \delta_{c'} - \delta_c \geq 1$). Per-window temporal centering over $L = 32$ leaves a small finite-window residual at $\tau = 0$, but the drift energy is concentrated at $\tau \geq 1$ (Fig. 2c). The benchmark is therefore the cleanest possible separator: the commutative monitors we evaluate — raw-level, instantaneous-covariance, and the periodic spectral-truncation kernel — are all at chance and the drift is exposed only by the causal noncommutative ($n \geq 2$) kernel, which is the empirical content of Theorem 2.

c. Causal spectral-truncation kernel. A window $X \in \mathbb{R}^{L \times C}$ ($L = 32$, $C = 7$) is centered in time per channel and mapped to the feature tensor $\Phi_n(X)_{cc'\tau} = \langle X_{:,c}, S^\tau X_{:,c'} \rangle / (L - \tau)$ for $\tau = 0, \dots, n - 1$, with S the causal lower shift; the operator-valued kernel is (1) and the scalar kernel is the Hilbert–Schmidt inner product of these features. The commutative member ($n=1$) retains only the instantaneous covariance, and each $n > 1$ adds one band of ordered lagged cross-channel coor-

dinates. For comparison we implement the *periodic* spectral-truncation baseline unchanged: it maps each channel to the $n \times n$ Toeplitz matrix of its truncated Fourier coefficients and forms the ordered cross-channel products of those matrices, exactly the construction of Hashimoto et al. [9]. The two kernels differ only in whether the truncated algebra is the causal shift (ours) or the periodic Fourier multiplier (theirs), which isolates causality as the operative ingredient.

d. Forecasting. We compare persistence (carry the last T_1 forward), climatology (the training-mean horizon trajectory), an autoregressive ridge on the 32-step T_1 history, and a multivariate ridge [64] on the flattened 32×7 window (ridge penalty $\alpha = 5$; this is the commutative $n=1$ kernel member), under a leak-free per-qubit 80/20 chronological split, scoring MAE per horizon step over five generator seeds with skill $1 - \text{MAE}/\text{MAE}_{\text{persistence}}$. The CST forecaster augments the window with the order- n causal features before the ridge solve; its eight-step skill versus n (Fig. 3c) is averaged over the same five seeds. The chronological split guarantees that no future sample informs a past prediction.

e. Detection. The unsupervised detector standardizes the kernel feature map on nominal training windows, fits a rank- k principal subspace by principal component analysis [65], and scores each test window by its mean squared reconstruction residual outside that subspace. Detection quality is ROC-AUC against the drift labels over eight randomized 70/30 splits; because the detector is fit on nominal windows only, randomization cannot leak a label it never uses. The truncation sweep reports, per order n , the best ROC-AUC over bottleneck ranks $k \in \{1, 2, 3, 5, 8, 12, 16\}$, and the kernel comparison (Table II) fixes $k = 8$ so that all detectors are compared at matched representational budget.

f. Computational complexity and resource accounting. Each window $X \in \mathbb{R}^{L \times C}$ is mapped to a feature tensor $\Phi_n(X)$ of nC^2 coordinates, every coordinate an inner product of length at most L , so feature extraction costs $O(nC^2L)$ time and $O(nC^2)$ memory per window — linear in the window length and the truncation order, and quadratic only in the channel count C (here $C=7$, $L=32$, and $n \leq 16$, for at most 784 feature coordinates). Fitting the detector standardizes these features and extracts a rank- k principal subspace of the $N_{\text{train}} \times nC^2$ nominal feature matrix, an $O(N_{\text{train}} nC^2 \min(N_{\text{train}}, nC^2))$ singular-value decomposition; scoring a test window then costs $O(nC^2L)$ to featurize and $O(nC^2k)$ to project onto the subspace and measure the residual. Every step is a dense linear-algebra primitive with no iterative optimization, no gradients, and no learned parameters, so the entire truncation-and-rank sweep is polynomial and runs in seconds on a single CPU core. This is the resource counterpart of the operating-characteristic result: the monitor is training-free at inference, and its cost grows only linearly in the noncommutativity order n that it is permitted to exploit.

g. Reproducibility and compute. The analyses run entirely on a single CPU and regenerate in seconds, with no GPU and no training. Every figure and table is computed live from the installed `qdriftforecast` package by `make_paper_figures.py` (the installed `qdrift-reproduce` entry point), and reported uncertainties are mean $\pm$ standard deviation across the $n = 5$ generator seeds (forecasting) or $n = 8$ randomized splits (detection). Determinism is byte-level: the seeds and the build epoch are pinned and the vector PDF output is fixed across runs, so every number in the manuscript is a reproducible function of integer seeds.

DATA AVAILABILITY

The synthetic multi-qubit telemetry and the correlated-decoherence benchmark used in this study are generated deterministically by the released code from fixed integer seeds and require no external download. The exact metric values underlying every figure and table are reported in the manuscript and are reproducible from the released code.

CODE AVAILABILITY

The telemetry generator, the causal spectral-truncation kernel, the detection and forecasting modules, and the figure-and-table pipeline are openly available in the project repository under an MIT license as the installable `qdriftforecast` package. All figures, tables, and reported uncertainties regenerate from the released code on commodity CPU hardware in seconds.

ACKNOWLEDGMENTS

The author thanks the open-source scientific-Python community. The author declares no competing interests.

Appendix A: Extended data tables

Appendix B: Supplementary derivations and reproduction details

This appendix is a self-contained derivation of every concept the released code implements, written for a reader with linear algebra and basic probability but no prior exposure to quantum-hardware telemetry, operator-algebraic kernels, or the estimators used here. Every symbol that appears in the code is defined, and each subsection names the source file (under `submission/code/`) that realizes it, so the appendix doubles as a guide to the artifact. The whole pipeline is CPU-only and deterministic: every number in the paper is a fixed function of integer seeds.

TABLE III. **Detection ROC-AUC versus truncation order.** Mean $\pm$ s.d. over the $n = 8$ randomized splits; best ROC-AUC over bottleneck ranks $k \in \{1, \dots, 16\}$ per order. Detection is at chance at the commutative order $n=1$, peaks sharply at $n=2$ (in **bold**), and decays monotonically toward chance as the representation becomes over-complete — the certified mechanism of Theorem 3 and the source data for Fig. 4a.

Order n	ROC-AUC	Order n	ROC-AUC
1	0.513 ± 0.041	6	0.692 ± 0.024
2	0.903 ± 0.016	8	0.647 ± 0.026
3	0.832 ± 0.022	10	0.619 ± 0.028
4	0.776 ± 0.023	12	0.596 ± 0.026
5	0.728 ± 0.023	16	0.573 ± 0.025

TABLE IV. **Per-horizon forecasting error.** Mean absolute error in μs , mean $\pm$ s.d. over the $n = 5$ telemetry-generator seeds. Each row is one step of the eight-step T_1 horizon, and the best (lowest) value per row is in **bold**. The learned ridge forecasters stay nearly flat across the horizon while persistence degrades monotonically and climatology is uniformly poor — the per-step view underlying Fig. 3a and the source of the widening skill margin.

Horizon (steps)	Persistence	Climatology	AR-ridge	Multivariate ridge
1	2.44 ± 0.16	13.01 ± 0.05	1.85 ± 0.07	1.80 ± 0.09
2	2.70 ± 0.11	12.89 ± 0.05	1.88 ± 0.08	1.82 ± 0.12
3	3.21 ± 0.08	12.79 ± 0.07	1.91 ± 0.05	1.84 ± 0.11
4	3.78 ± 0.17	12.69 ± 0.06	1.96 ± 0.07	1.84 ± 0.11
5	4.47 ± 0.11	12.63 ± 0.06	2.04 ± 0.06	1.88 ± 0.10
6	5.25 ± 0.19	12.55 ± 0.06	2.11 ± 0.05	1.91 ± 0.10
7	6.05 ± 0.14	12.49 ± 0.04	2.14 ± 0.07	1.90 ± 0.08
8	6.79 ± 0.18	12.40 ± 0.05	2.19 ± 0.06	1.88 ± 0.06

a. Source-file map. All paths are relative to `submission/code/`: `qdriftforecast/data.py` (telemetry generator, correlated-drift benchmark, windowing, splitting); `qdriftforecast/kernels.py` (the causal spectral-truncation kernel, its operator- and scalar-valued forms, and the periodic baseline); `qdriftforecast/detection.py` (the reconstruction detector, truncation sweep, (n, k) grid and kernel comparison); `qdriftforecast/forecasting.py` (forecasters); `make_paper_figures.py` (all figures and table bodies); and `generated_data/*.json` (the numbers behind each panel).

S2. Computation model: telemetry as arrays

(*qdriftforecast/data.py.*) A telemetry record is a finite table in which each row is a `(timestamp, qubit_id)` pair holding measured or simulated features. The seven channels (`FEATURE_COLS`) are `T1_us`, `T2_us`, `gate_fidelity_1q`, `gate_fidelity_2q`, `readout_error`, `gate_error_per_clifford`, and `cross_resonance_phase_rad`. Every result in the paper is a deterministic function of these arrays and fixed integer seeds.

S1. The problem in one paragraph

A quantum processor is not static: its coherence times, gate fidelities, readout error, and coupling parameters drift over hours, eroding the error budget that quantum error correction depends on, so an operator must decide *when* a device has drifted and *which* intervals warrant recalibration. We pose two tasks on the multivariate telemetry — *forecast* near-future coherence, and *detect* drifted windows without labels — and the central finding concerns the second: one geometry of drift, correlated decoherence living in ordered, lagged cross-channel structure, leaves every commutative monitor we evaluate at chance and is detectable only by a noncommutative one.

S3. The physics of the generator

(*generate_synthetic_dataset.*) For qubit q at time t the relaxation time is $T_1(q, t) = \max(10, 80 + 20 \sin(2\pi t/48 + 0.8q) - 0.05t + \varepsilon)$ with $\varepsilon \sim \mathcal{N}(0, 2^2)$, superposing a 48-hour oscillation (recalibration-like cycling, with phase $0.8q$ decorrelating the qubits), a secular decay (aging), and Gaussian fluctuation. The remaining channels track T_1 by construction, and a binary drift label marks any step with $T_1 < 50 \mu\text{s}$ or single-qubit fidelity below 0.998. Because the deterministic part is smooth and the noise small, the detrended series is weakly stationary with a slowly decaying autocorrelation — the structure that S8 converts into forecast skill.

S4. The correlated-decoherence benchmark, and why commutative monitors are at chance

(*generate_correlated_drift_windows.*) Both classes share the same base trajectory B and noise scales σ . The drift class adds a shared white latent u to four channels at delays $\delta_c \in \{0, 1, 2, 3\}$ under variance-preserving mixing $x_c = B_c + \sigma_c(\sqrt{1-a^2}\varepsilon_c + a u_{t-\delta_c})$. Because u is white, three matchings hold exactly: (i) each channel’s marginal variance is σ_c^2 in both classes; (ii) each channel’s temporal autocorrelation is unchanged, since a sum of independent white processes is white; and (iii) the *population* lag-0 cross-covariance between two injected channels is $a^2\sigma_c\sigma_{c'}\mathbb{E}[u_{t-\delta_c}u_{t-\delta_{c'}}] = 0$ for $\delta_c \neq \delta_{c'}$, matching nominal. The signal energy resides entirely in the lag- τ cross-covariance with $\tau = \delta_{c'} - \delta_c \geq 1$. Two caveats bound the scope of this matching: per-window temporal centering over the finite $L = 32$ window leaves a small $O(1/L)$ residual in the centered lag-0 cross-covariance of the drift class, so the matching is exact only in the stationary limit; and Theorem 2 is a statement about *first-moment* discriminability, not about every conceivable nonlinear statistic. The operative claim is the one we verify empirically: the commutative monitors we evaluate — the raw-level detector, the commutative ($n=1$) kernel, and the periodic spectral-truncation kernel — are all at chance ($n=1$ ROC-AUC 0.51), because the drift energy is concentrated in the ordered, lagged cross-channel correlations they cannot represent (Fig. 2c).

S5. The causal spectral-truncation kernel, from scratch

(*qdriftforecast/kernels.py.*) Center each channel of a window in time. For lags $\tau = 0, \dots, n-1$ form the ordered cross-channel correlation $\Phi_n(X)_{cc'\tau} = \frac{1}{L-\tau} \sum_{t \geq \tau} X_{t,c} X_{t-\tau,c'}$, which are the matrix elements of $X^\top S^\tau X$ with S the causal lower shift. The operator-valued kernel $K_n(X, Y) = \sum_{\tau < n} w_\tau X^\top S^\tau Y \in M_C$ is an element of the channel C^* -algebra, and the scalar kernel is $\langle \Phi_n(X), \Phi_n(Y) \rangle$. At $n=1$ this is the symmetric instantaneous covariance and is commutative; for $n \geq 2$ the lagged terms are asymmetric in (c, c') because $S^\top \neq S$, which is exactly the noncommutativity the result turns on. Truncating the shift algebra to n powers is the causal analog of truncating the Fourier-multiplier algebra to n modes in the spectral-truncation construction of Hashimoto et al., and the periodic baseline (*periodic_truncation_feature_map*) implements that original torus construction unchanged, so the comparison in Fig. 4c varies only causality.

S6. The detector and the optimal-truncation certificate

(*qdriftforecast/detection.py.*) Standardize the kernel features on nominal windows, fit a rank- k principal subspace, and score a test window by its residual energy outside it. Increasing the truncation order n simultaneously captures more drift signal (good) and admits C^2 noise coordinates per lag (bad), so the separation-to-noise ratio is unimodal in n , peaking just past the informative lag (Theorem 3). Empirically the peak is at $n=2$ with ROC-AUC 0.90; the commutative endpoint and the over-complete limit are both at chance, and the rank axis of the same certificate recovers the classical low-rank over-completeness collapse.

S7. Metrics: MAE for forecasting, ROC-AUC for detection

Forecasting uses mean absolute error per horizon step, reported step by step so the widening gap over persistence is visible, with skill $1 - \text{MAE}/\text{MAE}_{\text{pers}}$. Detection uses ROC-AUC, which by the Mann–Whitney identity (Proposition 4) equals the probability that a random drift window outranks a random nominal one, is invariant under per-feature rescaling, and equals $\frac{1}{2}$ for a constant score — the precise chance level that the commutative and over-complete monitors attain.

S8. Why forecast skill grows with the horizon

This is Theorem 5. For a weakly stationary process with autocorrelation $\rho(h)$, the persistence risk is $2\gamma_0(1 - \rho(h))$ and the best single-lag predictor’s risk is $\gamma_0(1 - \rho(h)^2)$, so the skill is $(1 - \rho(h))/2$, which increases as ρ decays. The multivariate ridge uses the full history and exceeds this single-lag bound, reaching 72% at horizon eight, while the noncommutative coordinates add nothing because the marginal structure is already linear and instantaneous (S5).

S9. How to reproduce everything

Install with `pip install qdriftforecast` (or, from a checkout, `pip install submission/code`). Run `qdrift-reproduce` (equivalently `python -m qdriftforecast.reproduce` or `python make_paper_figures.py` from `submission/code/`). The script writes, deterministically from fixed seeds on a single CPU: the five vector PDF figures into `./figures/`; the three table bodies into `./tables/` (`forecasting_benchmark.tex`, `detection_benchmark.tex`, `truncation_sweep.tex`); and the machine-readable source data into `generated_data/*.json`. No number in the manuscript is typed by hand; each is emitted by this pipeline.

S10. What the result means, and what it does not

A small, training-free, noncommutative kernel detects a correlated-decoherence drift that every commutative monitor — including the periodic spectral-truncation kernel — misses entirely, and a single truncation order certifies how much noncommutativity to use. Coherence forecasting succeeds wherever autocorrelation persists, and there noncommutativity is unnecessary. We make

no quantum-hardware superiority claim: the telemetry is a physically motivated model and the correlated-decoherence mode is a controlled construction chosen to isolate the noncommutative effect. What the work delivers is a reusable, leakage-controlled, CPU-reproducible, operator-algebraic foundation for quantum-processor reliability monitoring, together with a demonstration — both empirical and certified — that the geometry of the drift dictates the noncommutativity a monitor must have.

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